



Introduction to CFD (4RC30)

**Role of pressure;
solving PDE's**

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Outline

Summary of last lectures

Content of this lecture

- Chapter 6
 - Role of pressure; staggered grid
 - Momentum equations
 - SIMPLE algorithm
- Chapter 7
 - Solving the discretized equations
 - Tri-diagonal matrix algorithm
- Examples

Wrap up

Summary of last lectures

- Rate of change + convection = diffusion + (source – sink)

$$\frac{\partial(\rho\phi)}{\partial t} + \text{div}(\rho\phi\mathbf{u}) = \text{div}(\Gamma \text{grad } \phi) + S_\phi$$

- General form of discretized linear equations:

$$\boxed{a_P \phi_P = \sum a_{nb} \phi_{nb} + S_u} \quad \boxed{a_P = \sum a_{nb} - S_P}$$

Source terms are included through S_u and S_P

- Spatial discretization schemes: CDS, UDS and Hybrid

Source term linearization

- **Purpose:** to solve the discretized equations in a linear framework
- The source term is expressed in a linear form: $\bar{S}\Delta V = S_u + S_P\phi_P$
- $\bar{S}$ is the volume-averaged value, S_u the constant part, $S_P(\leq 0)$ the linear component
- **Taylor series expansion:**

$$S^{k+1} = S^k + \left(\frac{\partial S}{\partial \phi}\right)^k (\phi_P^{k+1} - \phi_P^k) = \underbrace{\left(S - \left(\frac{\partial S}{\partial \phi}\right) \phi_P\right)^k}_{S_u/\Delta V} + \underbrace{\left(\frac{\partial S}{\partial \phi}\right)^k}_{S_P/\Delta V} \phi_P^{k+1}$$

old step
new step

Examples:

	S_u	S_P	$S_u/\Delta V$	$S_P/\Delta V$
uniform	$S\Delta V = q$	$q\Delta V$	0	
variable-dependent	$S\Delta V = C\phi + q$	$q\Delta V$	$C\Delta V$	
variable-dependent	$S\Delta V = -\phi^3 + q$	$(2\phi_P^{k^3} + q)\Delta V$		$-3\phi_P^{k^2}\Delta V$

Role of pressure

- **Complete Navier-Stokes equations:**

$$\frac{\partial(\rho u)}{\partial t} + \text{div}(\rho u \mathbf{u}) = -\frac{\partial p}{\partial x} + \text{div}(\mu \text{grad } u) + S_{Mx}$$

$$\frac{\partial(\rho v)}{\partial t} + \text{div}(\rho v \mathbf{u}) = -\frac{\partial p}{\partial y} + \text{div}(\mu \text{grad } v) + S_{My}$$

- **Steady state situation ($\partial/\partial t = 0$):**

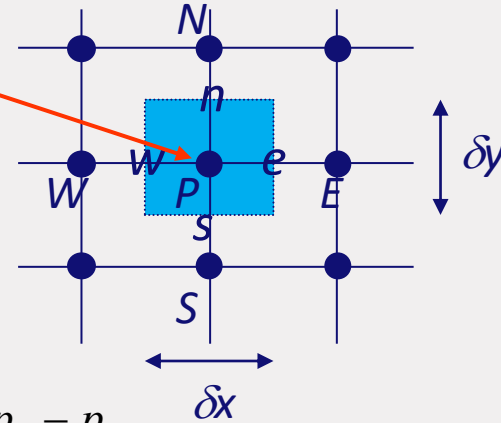
$$\text{div}(\rho u \mathbf{u}) = -\frac{\partial p}{\partial x} + \text{div}(\mu \text{grad } u) + S_{Mx}$$

$$\text{div}(\rho v \mathbf{u}) = -\frac{\partial p}{\partial y} + \text{div}(\mu \text{grad } v) + S_{My}$$

- **Pressure gradient often driving force flow**

Collocated grid

- Pressure and velocities stored on cell centers

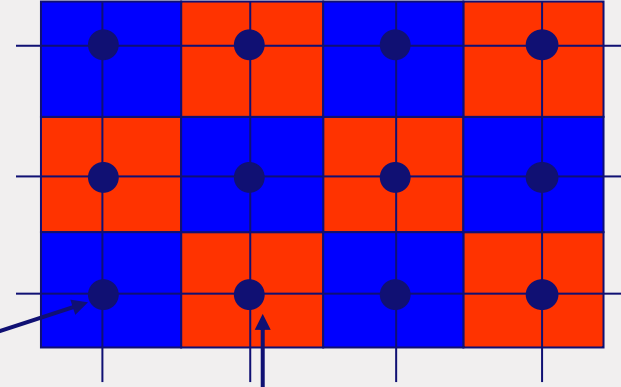
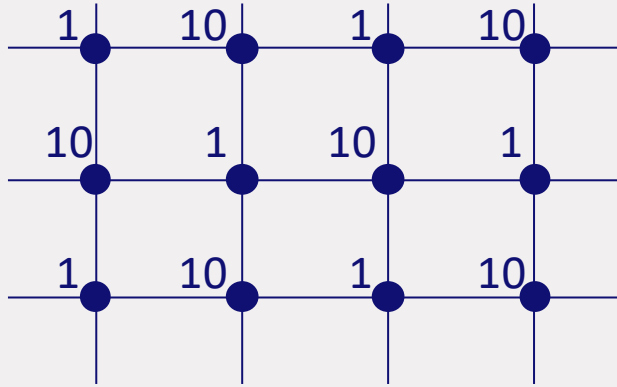


$$\left(\frac{\partial p}{\partial x} \right)_P = \frac{p_e - p_w}{\delta x} = \frac{1}{\delta x} \left\{ \left(\frac{p_E + p_P}{2} \right) - \left(\frac{p_P + p_W}{2} \right) \right\} = \frac{p_E - p_W}{2\delta x}$$

$$\left(\frac{\partial p}{\partial y} \right)_P = \frac{p_N - p_S}{2\delta y}$$

Pressure at P doesn't appear in gradients!

'Checkerboard' pressure field



$$\left(\frac{\partial p}{\partial x} \right)_P = \frac{p_E - p_W}{2\delta x} = 0$$

$$\left(\frac{\partial p}{\partial y} \right)_P = \frac{p_N - p_S}{2\delta y} = 0$$

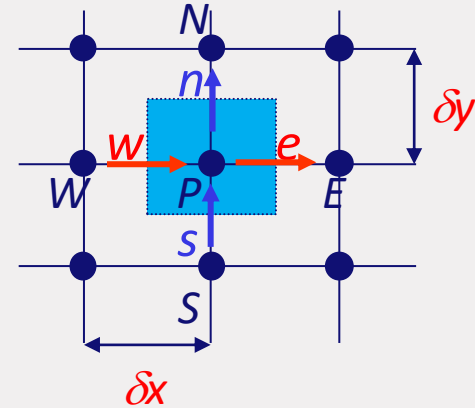
$$\left(\frac{\partial p}{\partial x} \right)_P = \frac{p_E - p_W}{2\delta x} = 0$$

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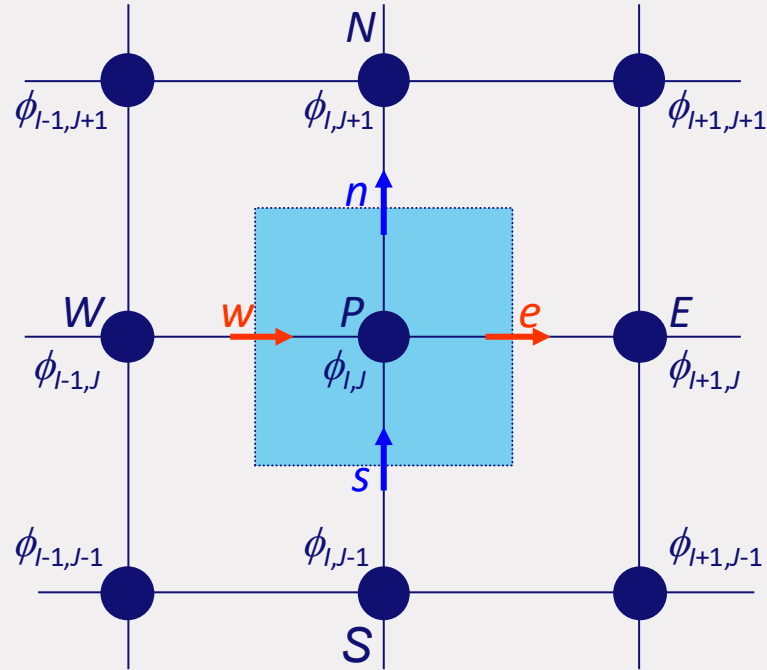
Solution: staggered grid

- Store pressure at cell centers (P_W, P_P, P_E , etc.)
- Store velocities at cell faces (u_w, u_e, v_s, v_n)
- Advantages:
 - No checkerboard pressure field possible
 - Velocities are stored at desired position

$$\left(\frac{\partial p}{\partial x}\right)_w = \frac{p_P - p_W}{\delta x_u} \quad \left(\frac{\partial p}{\partial y}\right)_s = \frac{p_P - p_S}{\delta y_v}$$
$$\left(\frac{\partial p}{\partial x}\right)_e = \frac{p_E - p_P}{\delta x_u} \quad \left(\frac{\partial p}{\partial y}\right)_n = \frac{p_N - p_P}{\delta y_v}$$

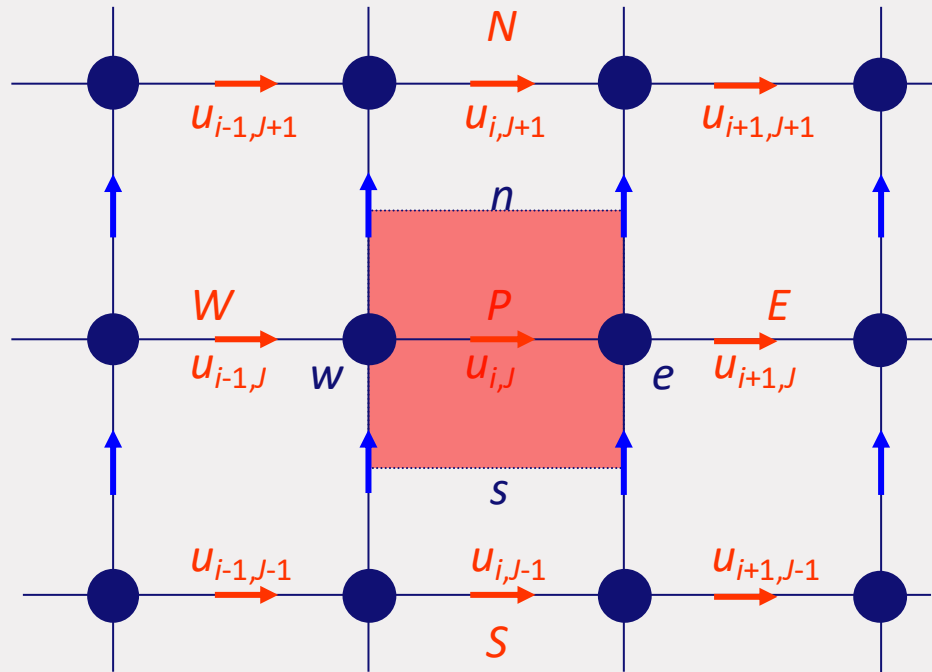


Staggered grid: numbering $\phi_{I,J}$



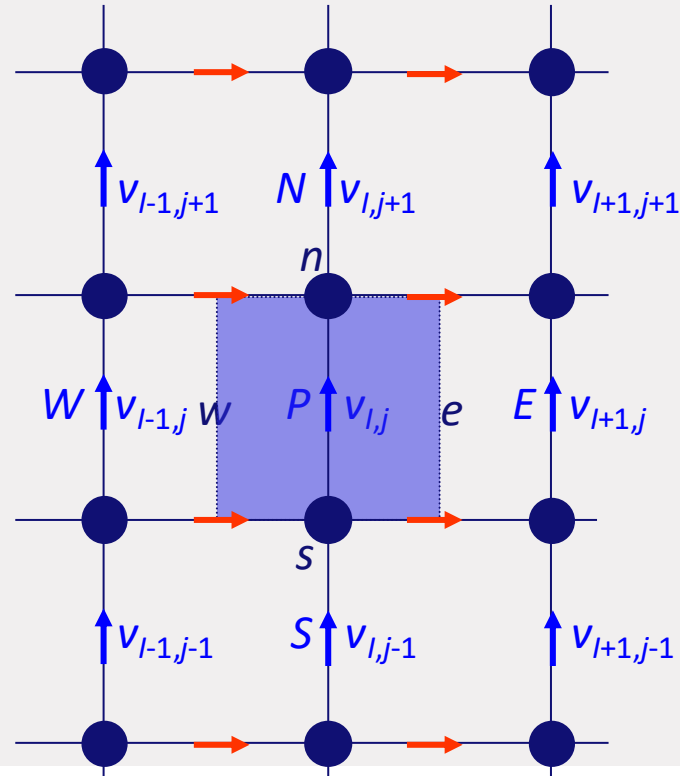
(Fig. 6.5)

Staggered grid: numbering $u_{i,j}$



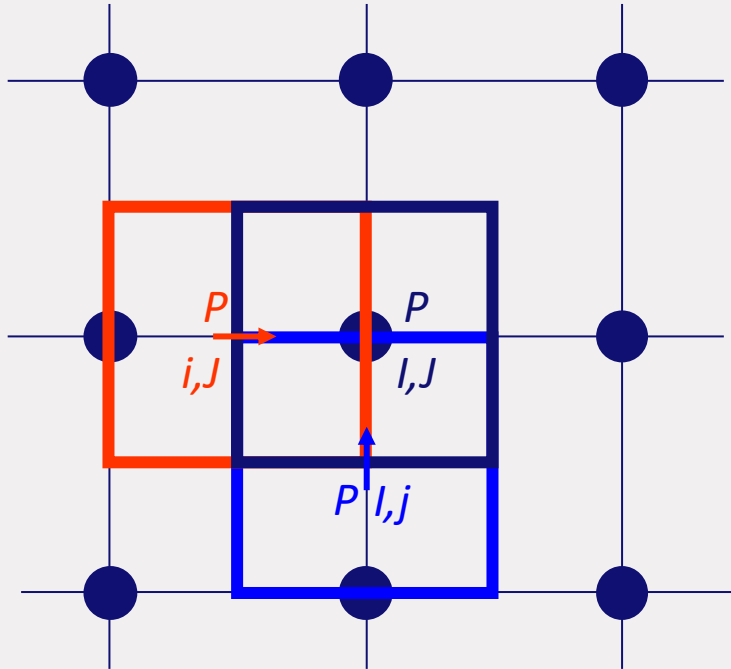
(Fig. 6.3)

Staggered grid: numbering $v_{I,j}$



(Fig. 6.4)

Staggered grid: overview



(Fig. 6.2)

```
Dx = XMAX/NPI;
```

```
% Length variable for the scalar points in the x direction
```

```
x(1) = 0.;
```

```
x(2) = 0.5*Dx;
```

```
for I = 3:NPI+1
```

```
    x(I) = x(I-1) + Dx;
```

```
end
```

```
x(NPI+2) = x(NPI+1) + 0.5*Dx;
```

```
% Length variable for the velocity components u(i,j) in the x direction
```

```
x_u(1) = 0.;
```

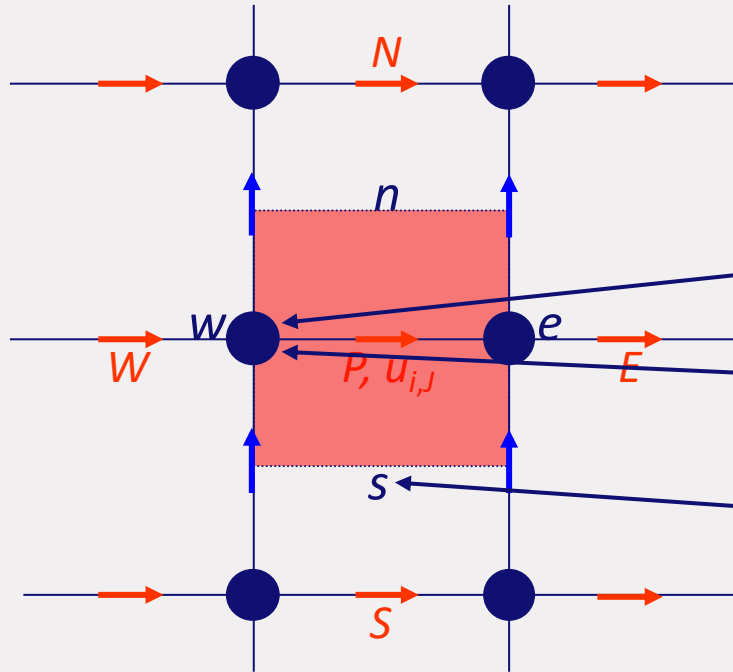
```
x_u(2) = 0.;
```

```
for i = 3:NPI+2
```

```
    x_u(i) = x_u(i-1) + Dx;
```

```
end
```

Momentum equations



$$a_P u_P = \sum a_{nb} u_{nb} + (p_w - p_e) A_P + b$$

flow + diffusion
forces

pressure
force

other body
forces

$$F_w = (\rho u)_w = \frac{F_{i,j} + F_{i-1,j}}{2}$$

$$D_w = \Gamma_w / \delta x = \frac{\Gamma_{i-1,j}}{x_i - x_{i-1}}$$

$$D_s = \Gamma_s / \delta y_v$$

$$= \frac{\Gamma_{i-1,j} + \Gamma_{i,j} + \Gamma_{i-1,j-1} + \Gamma_{i,j-1}}{4(y_j - y_{j-1})}$$

Momentum equations

$$a_P u_P = \sum a_{nb} u_{nb} + (p_w - p_e) A_P + b$$

$$a_P v_P = \sum a_{nb} v_{nb} + (p_s - p_n) A_P + b$$

- **Values of a_{nb} are function of F and D :**
 - F and D depend on discretization scheme
 - F and D need to be interpolated to cell faces
- **Pressure gradient needs no interpolation**
- **Now we need to solve the u and v equations, using special treatment of p**

Solving the u equation

$$a_P u_P = \sum a_{nb} u_{nb} + (p_w - p_e) A_P + b$$

- **Guess u, v and p : u^*, v^* and p^***

$$a_P u_P^* = \sum a_{nb} u_{nb}^* + (p_w^* - p_e^*) A_P + b$$

- **Define pressure and velocity corrections:**

$$\begin{aligned} p &= p^* + p' & u &= u^* + u' & v &= v^* + v' \\ p' &= p - p^* & u' &= u - u^* & v' &= v - v^* \end{aligned}$$

- **Subtract guessed from correct velocity eqs.:**

$$a_P u'_P = \sum a_{nb} u'_{nb} + (p'_w - p'_e) A_P$$

Solving the u equation

$$a_P u'_P = \sum a_{nb} u'_{nb} + (p'_w - p'_e) A_P$$

- Omit correction **terms** of neighbors to obtain the velocity correction equation:

$$u'_P = d_P (p'_w - p'_e) \quad d_P \equiv A_P / a_P$$

- Now we know how to correct the guessed u^* :

$$u_P = u^*_P + d_P (p'_w - p'_e)$$

Solving the v equation

- We derived an expression for u' :

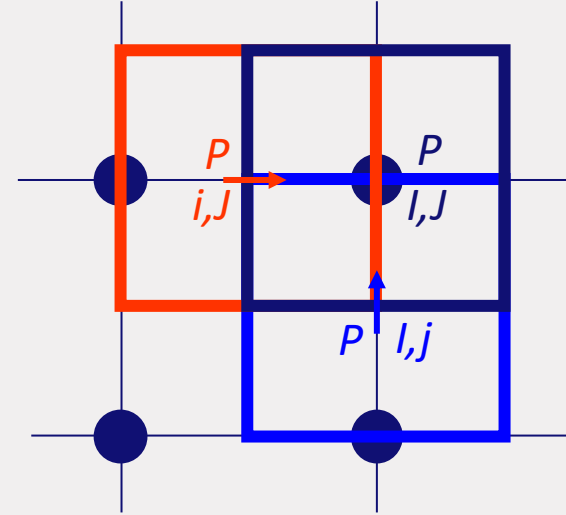
$$u_P = u_P^* + d_P(p'_w - p'_e)$$

- Analogous derivation gives expression for v' :

$$v_P = v_P^* + d_P(p'_s - p'_n)$$

$$d_P \equiv A_P / a_P$$

- Now we need an expression for p'



Pressure correction equation

- We use the continuity equation:

$$\text{div}(\rho \mathbf{u}) = 0 \Rightarrow [(\rho u)_e - (\rho u)_w] + [(\rho v)_n - (\rho v)_s] = 0$$

- Substitute $u = u^* + u'$ and $v = v^* + v'$:

$$\begin{aligned} &(\rho A)_e (u^*_e + d_e(p'_P - p'_E)) - \\ &(\rho A)_w (u^*_w + d_w(p'_W - p'_P)) + \\ &(\rho A)_n (v^*_n + d_n(p'_P - p'_N)) - \\ &(\rho A)_s (v^*_s + d_s(p'_S - p'_P)) = 0 \end{aligned}$$

Pressure correction equation

- Rearrange p' equation to yield:

$$a_P p'_P = a_E p'_E + a_W p'_W + a_N p'_N + a_S p'_S + b$$
$$a_P p'_P = \sum a_{nb} p'_{nb} + b$$

with:

$$a_E = (\rho dA)_e$$

$$a_W = (\rho dA)_w$$

$$a_N = (\rho dA)_n$$

$$a_S = (\rho dA)_s$$

$$a_P = a_E + a_W + a_N + a_S$$

$$b = (\rho u^* A)_e - (\rho u^* A)_w + (\rho v^* A)_s - (\rho v^* A)_n$$

Under-relaxation

- To prevent from divergence we use only part of the corrections:

$$\begin{aligned} p &= p^* + p' & p^n &= p^{n-1} + \alpha_p p' \\ u &= u^* + u' & u^n &= \alpha_u (u^* + u') + (1 - \alpha_u) u^{n-1} \Rightarrow u^n = u^{n-1} + \alpha_u u' \\ v &= v^* + v' & v^n &= \alpha_v (v^* + v') + (1 - \alpha_v) v^{n-1} \Rightarrow v^n = v^{n-1} + \alpha_v v' \end{aligned}$$

$$0 < \alpha_p, \alpha_u, \alpha_v \leq 1$$

Under-relaxation

$$a_P u_P = \sum a_{nb} u_{nb} + (p_w - p_e) A_P + b$$

$$a_P v_P = \sum a_{nb} v_{nb} + (p_s - p_n) A_P + b$$

- **Applying under-relaxation gives:**

$$\frac{a_P}{\alpha_u} u_P = \sum a_{nb} u_{nb} + (p_w - p_e) A_P + b + \left[(1 - \alpha_u) \frac{a_P}{\alpha_u} \right] u_P^{n-1}$$

$$\frac{a_P}{\alpha_v} v_P = \sum a_{nb} v_{nb} + (p_s - p_n) A_P + b + \left[(1 - \alpha_v) \frac{a_P}{\alpha_v} \right] v_P^{n-1}$$

SIMPLE

Semi-Implicit Method for Pressure-Linked Equations (SIMPLE)

- Solve discretized equations for u^* and v^*
- Solve discretized equation for p'
- Correct pressure and velocities
- Solve any other discretized equations for ϕ
- Convergence?
 - ✓ Yes: stop
 - ✓ No: set $p^* = p$, $u^* = u$, $v^* = v$, $\phi = \phi^*$
repeat sequence

Other algorithms linking u , v and p

- SIMPLER: SIMPLE Revised
- SIMPLEC: SIMPLE Consistent
- PISO: Pressure Implicit with Splitting of Operators
- See book Chapter 6 Versteeg and Malalasekera

Solution of discretized equations

$$a_P \phi_P = \sum a_{nb} \phi_{nb} + b$$

- **Gauss-Seidel point-by-point method**

$$\phi_P = \frac{\sum a_{nb} \phi_{nb} + b}{a_P}$$

- **Successive over-relaxation (SOR)**

$$\phi_P = \omega \frac{\sum a_{nb} \phi_{nb} + b}{a_P} - (\omega - 1) \phi_P^{\text{old}} \quad 1 < \omega < 2$$

Solution of discretized equations

- 1D: (n+2) equations and (n+1) unknowns

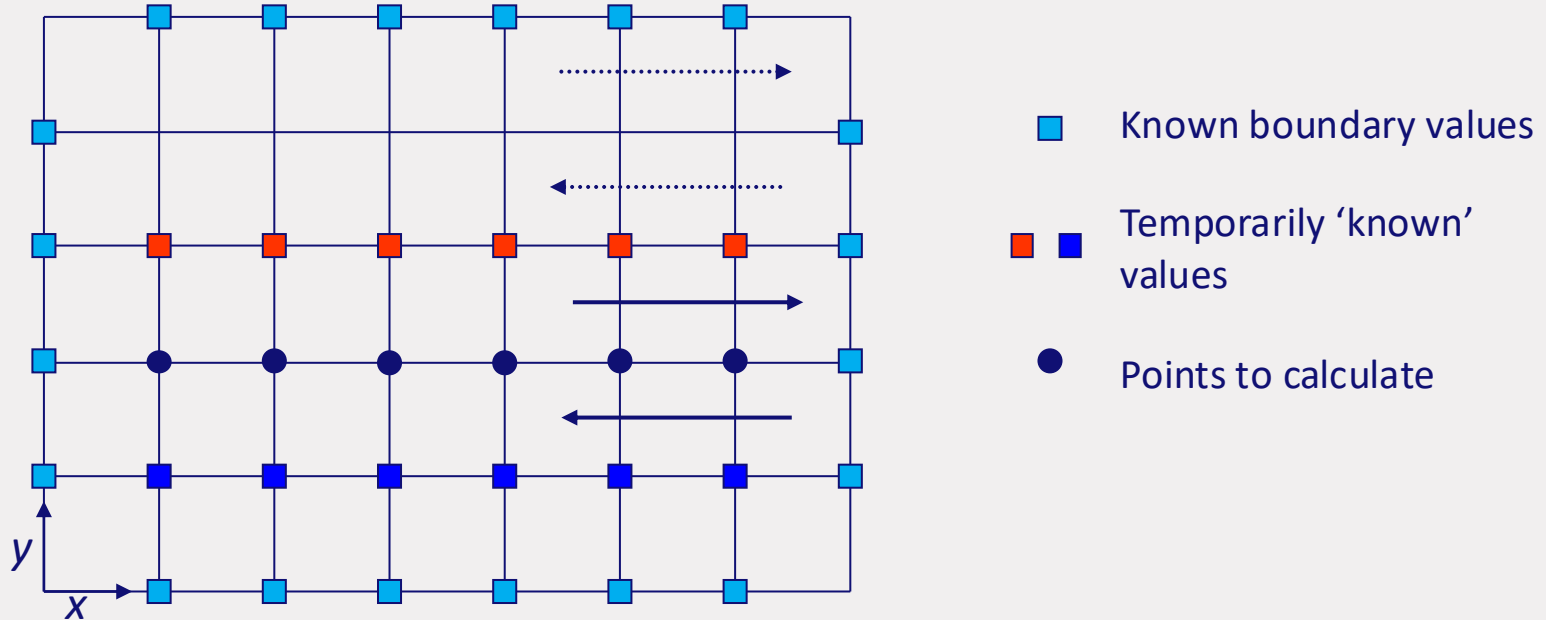
$$a_P \phi_P = \sum a_{nb} \phi_{nb} + b$$

$$\begin{array}{rcl}
 \phi_0 & & = b_0 \\
 -(a\phi)_0 + (a\phi)_1 - (a\phi)_2 & \swarrow & = b_1 \\
 -(a\phi)_1 + (a\phi)_2 - (a\phi)_3 & & = b_2 \\
 -(a\phi)_2 + (a\phi)_3 - (a\phi)_4 & \searrow & = b_3 \\
 -(a\phi)_{n-1} + (a\phi)_n - (a\phi)_{n+1} & & = b_n \\
 \phi_{n+1} & & = b_{n+1}
 \end{array}$$

- Solve by **forward elimination** and **backward substitution**
(Tri-diagonal matrix algorithm)

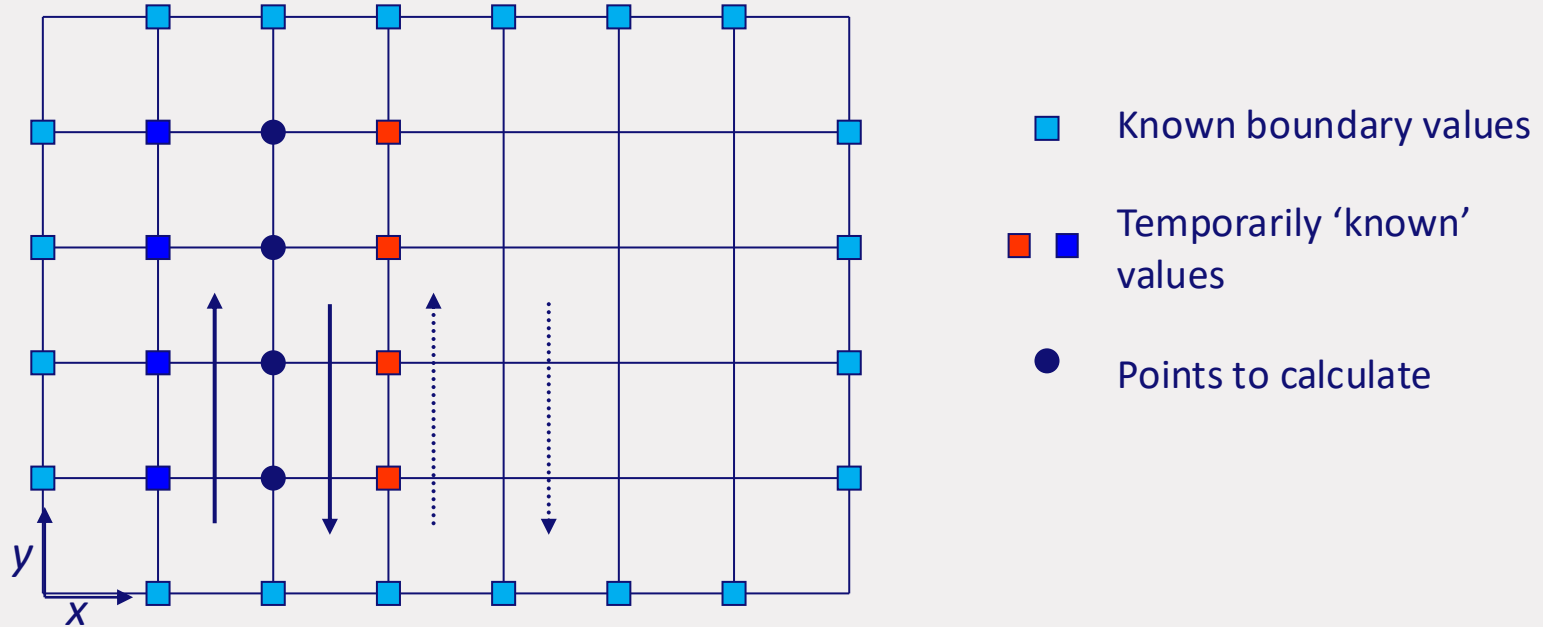
Solution of discretized equations

- 2D: apply TDMA line by line horizontally



Solution of discretized equations

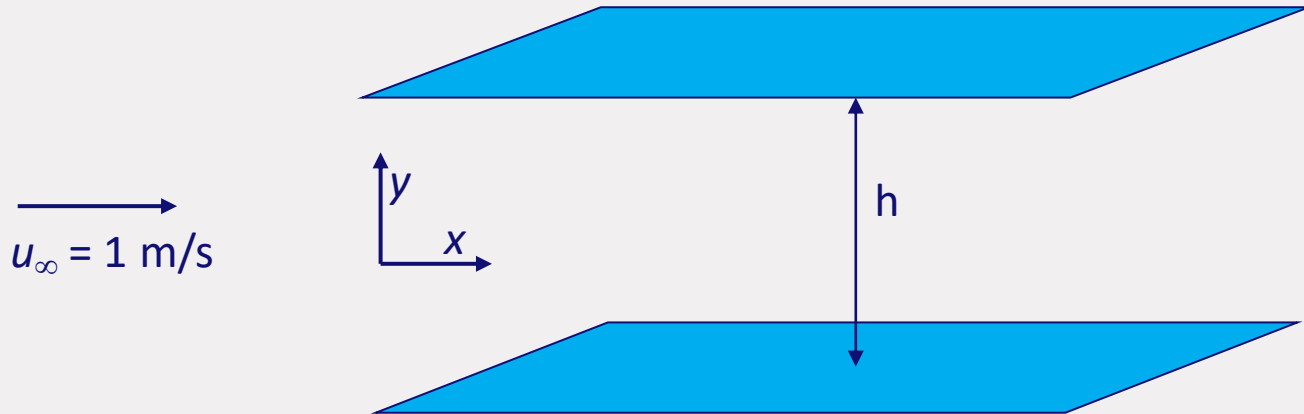
- 2D: apply TDMA line by line vertically



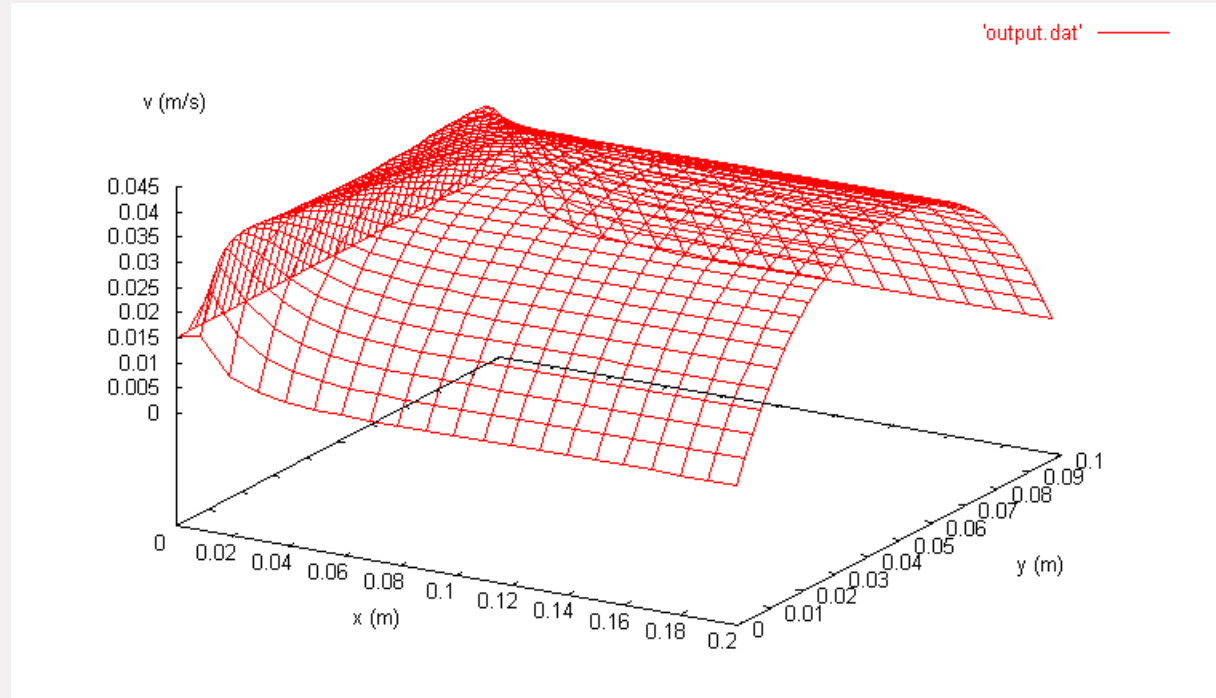
Flow between two parallel plates

- Analytical solution:

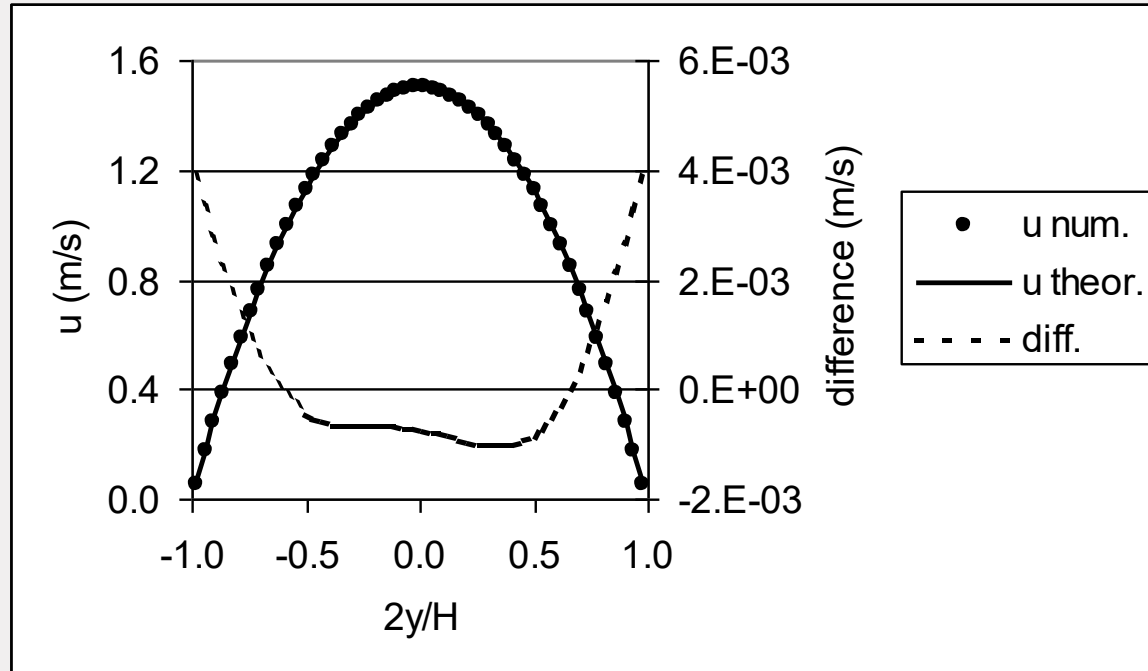
$$\frac{u(y)}{u_{\infty}} = \frac{3}{2} \left\{ 1 - \left(\frac{2y}{h} \right)^2 \right\}$$



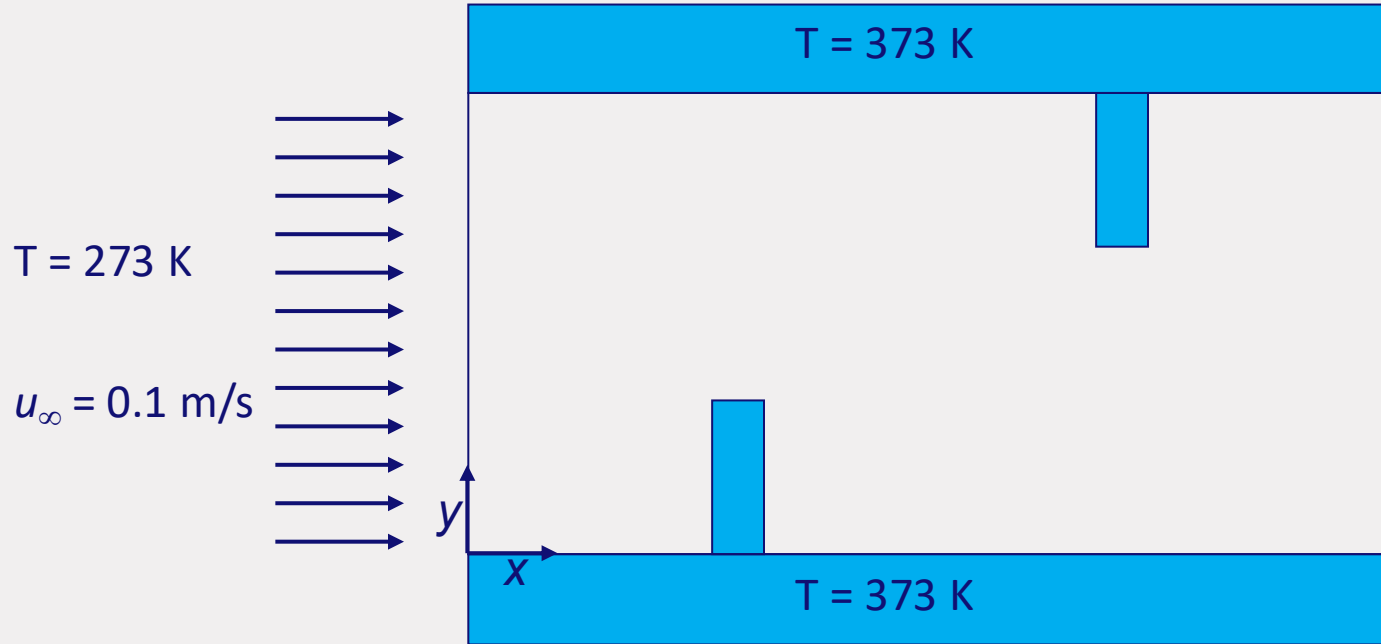
Flow between two parallel plates



Flow between two parallel plates



Flow between plates with baffles



Flow between plates with baffles

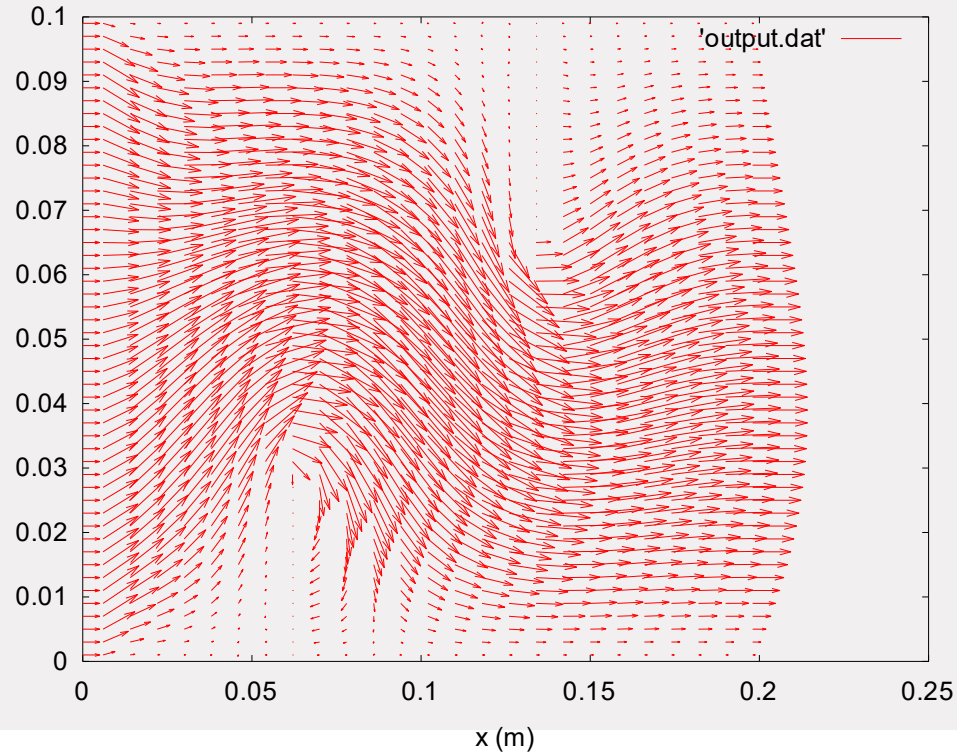
The governing equations are:

$$\operatorname{div}(\rho u \mathbf{u}) = -\frac{\partial p}{\partial x} + \operatorname{div}(\mu \operatorname{grad} u) + S_{Mx}$$

$$\operatorname{div}(\rho v \mathbf{u}) = -\frac{\partial p}{\partial y} + \operatorname{div}(\mu \operatorname{grad} v) + S_{My}$$

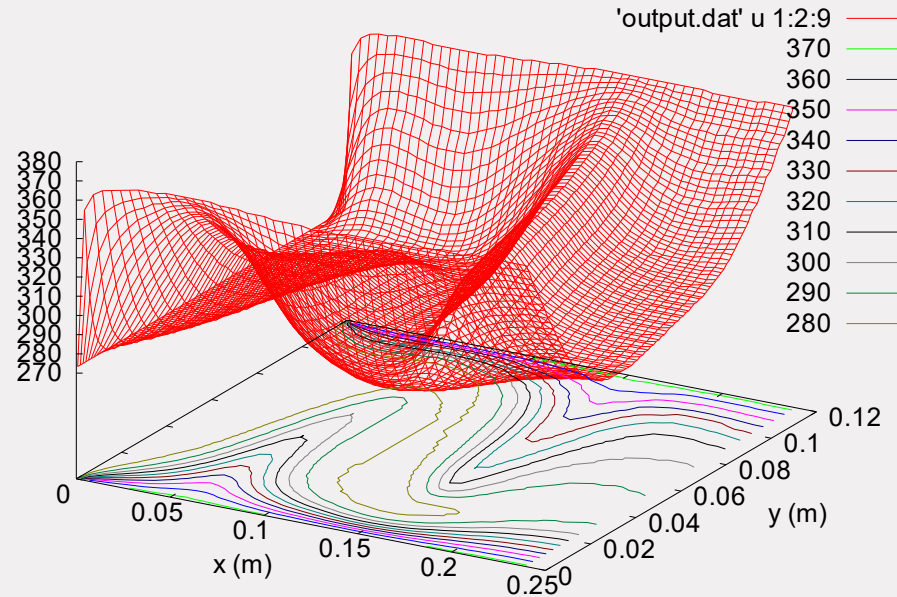
$$\operatorname{div}(\rho T \mathbf{u}) = \operatorname{div}\left(\frac{k}{C_p} \operatorname{grad} T\right)$$

Flow between plates with baffles



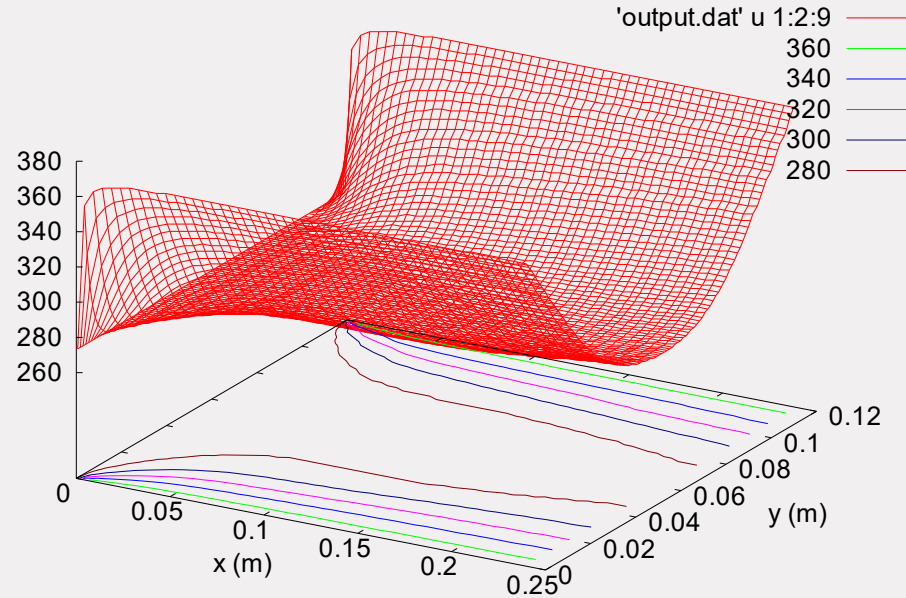
Flow between plates with baffles

Temperature distribution with baffles



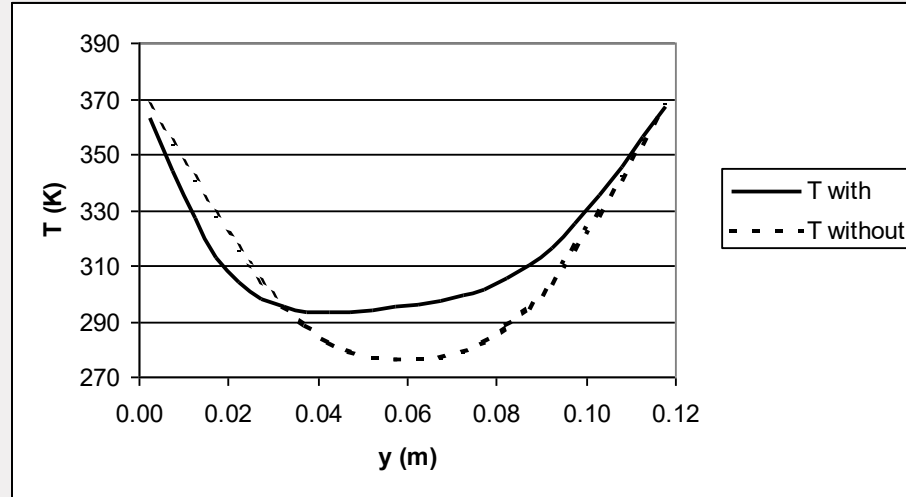
Flow between plates with baffles

Temperature distribution without baffles



Performance as heat exchanger

- With baffles $\Delta T = 42$ K
- Without baffles $\Delta T = 36$ K



Wrap up

- Source term linearization
- Role of pressure
- Staggered grid
- SIMPLE algorithm
- TDMA solver

Make groups

- 2 or 3 persons as a group for final assignment
- Make the group now and work together for tutorials
- Use “Partner wanted” post on Canvas